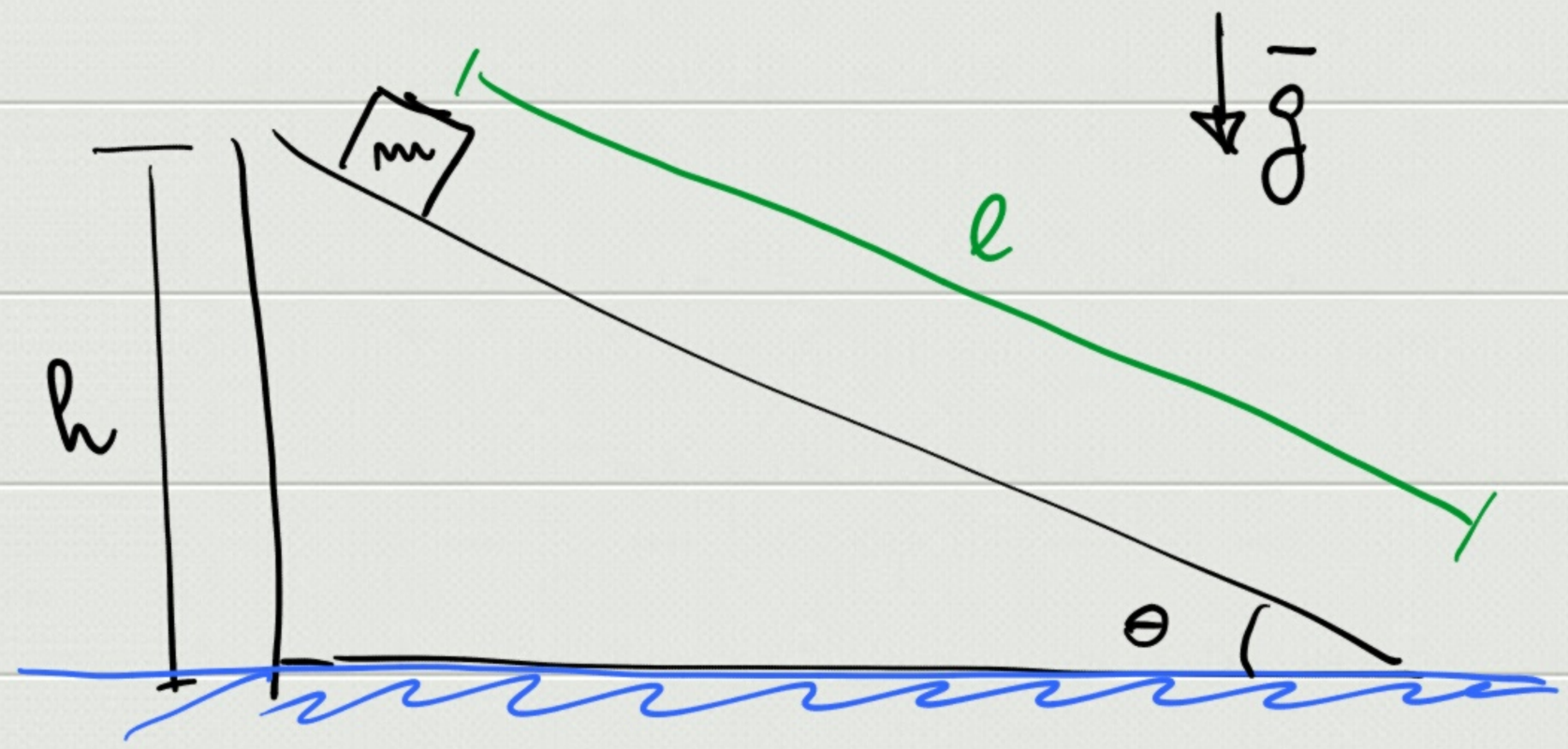
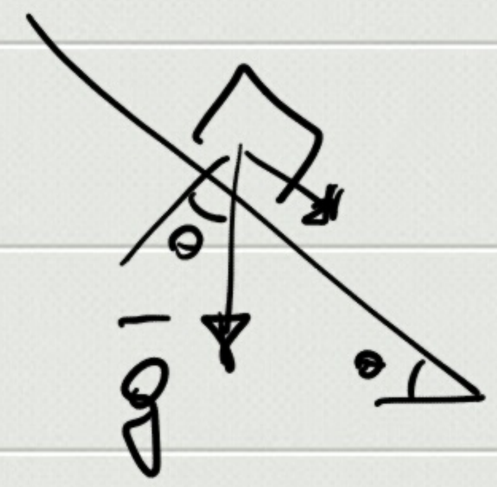



$m = 10\ kg$

$h = 0.5\ m$

$v_0 = 0$

$v_f = ?$ $t$ $(\theta = 30°, 60°)$




- $v^2 = 2g\sin\theta \cdot l$ * $\qquad v_f^2 = v_i^2 + 2a\Delta x$

- $\begin{cases} l = \frac{1}{2} g\sin\theta \cdot t^2 \\ v = g\sin\theta \cdot t \end{cases}$ *

- $\cancel{l = vt}$

- $mgh = \frac{1}{2} m v^2$ * $\qquad W_{peso} = -\Delta E_{p,peso} = \Delta E_k$

$$-(mgz_f - mgz_i)$$

$z_f \to 0$, $z_i \to h$

$$t = \sqrt{\frac{2l}{g\sin\theta}} = \sqrt{\frac{2h}{g\sin^2\theta}}$$

$h = l\sin\theta$

$l = \frac{h}{\sin\theta}$

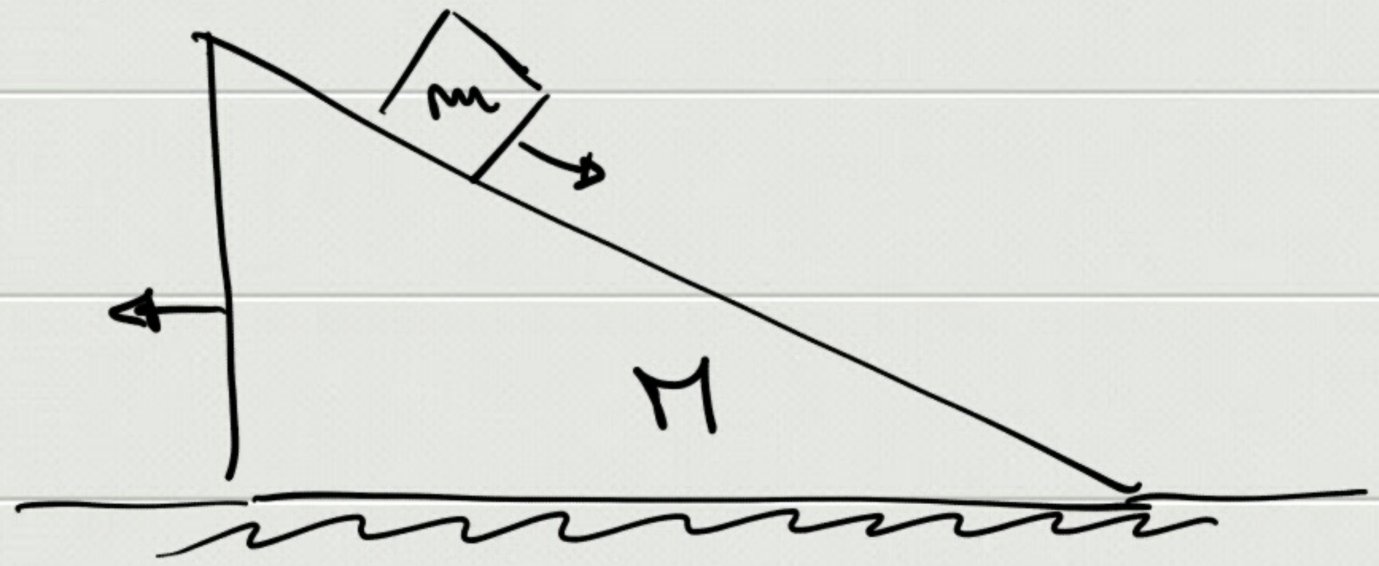
$t(\theta = 30°) = 0.41\ s$

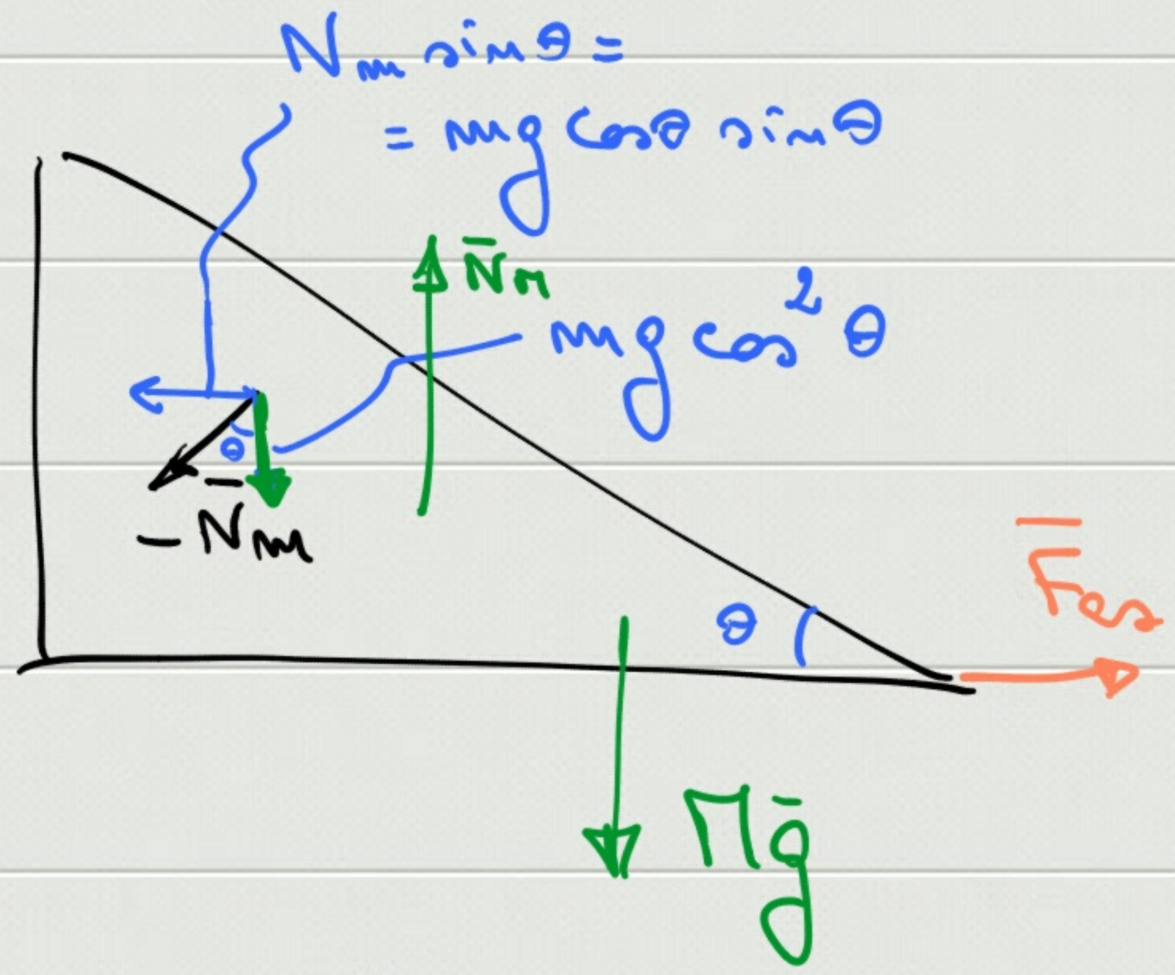
$t(\theta = 60°) = 0.14\ s$

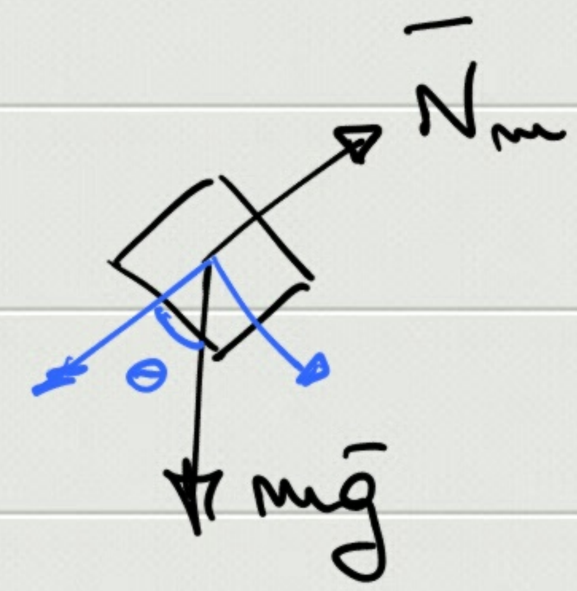
$$v = g\sin\theta \cdot t = g\sin\theta\sqrt{\frac{2h}{g\sin^2\theta}} = \sqrt{2gh}$$

$$= 3.13 \text{ m/s}$$

---




$$N_m = mg\cos\theta$$

$$N_M = Mg + mg\cos^2\theta$$

$$F_{es} = mg\sin\theta\cos\theta \le F_{es,max} = \mu_s N_M =$$

$$= \mu_s (M + m\cos^2\theta) g$$

$$\Rightarrow \boxed{\mu_s \geq \frac{m\sin\theta\cos\theta}{M + m\cos^2\theta}}$$